

67-262 Database Design and Development, Fall 2025

Project Phase-2 Report

Project Title: *Duolingo Database Development Project*

1. Updated List of User Stories

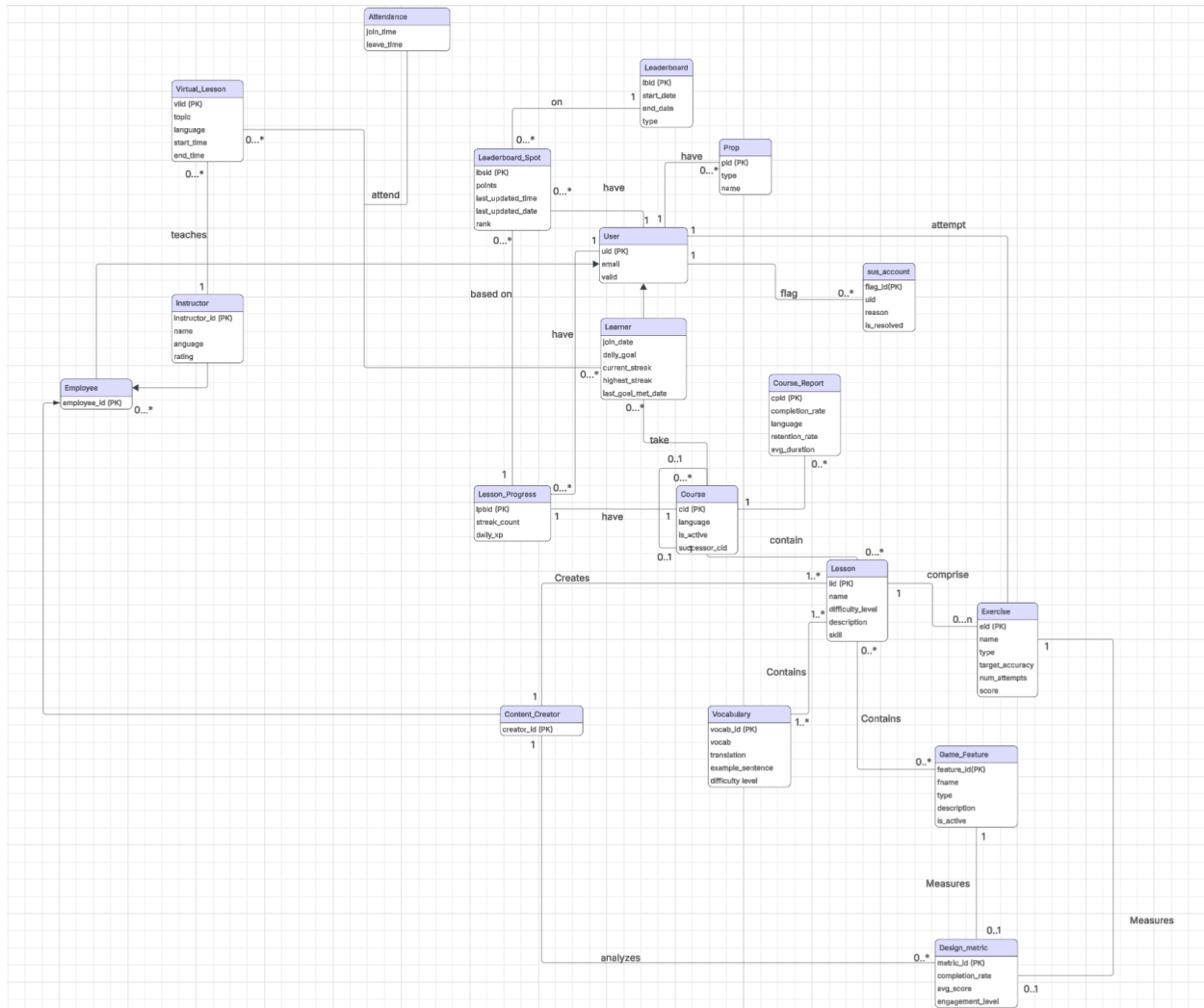
Based on feedback from TA on our Phase 1 Deliverable, we created a new user story 1 to fit the ERD and adjusted user story 7,8,9 as we were told the original user stories can't be implemented at the database level.

ID	Verb+Noun	As a <role>	I want <goal>	So that <reason>	Simple / Complex	Operational / Analytical
US1	Track props (boosters)	Language Learner	To keep track of my un-used props (streak-freezer etc)	I am constantly engaging with the gamification features of Duolingo	Simple	Operational
US2	See leaderboard	Language Learner	To see my rank against other learners on a leaderboard	Stay motivated to be consistent	Complex	Operational
US3	Track streaks/ daily goals	Language Learner	To track streaks and daily goals	So I don't lose progress, stay motivated and incline me to stay consistent	Complex	Operational
US4	View average accuracy	Content creator	To view a summary of the average accuracy rates for each exercise	So that I can identify which ones might need to be simplified or adjusted	Simple	Analytical
US5	Upload vocabulary	Content creator	To upload new vocabulary items with their corresponding translations and example sentences	So that learners can learn new words efficiently within the lesson	Simple	Operational

US6	Evaluate game-features	Content creator	To measure how different gamified elements affect lesson completion rates	So that I can determine which engagement tools are most effective.	Complex	Analytical
US7	Retire a course	Operation	To retire a language course and replace it with successor course	So learners always have access to the best-evaluated course offerings	Complex	Operational
US8	Flag suspicious account	Operation	To flag a user account as suspicious	So that our system can automatically restrict activity and prevent abuse (unreasonable progress).	Simple	Operational
US9	View average XP	Language Learner	To see my average daily XP earned across all lessons	So that I can understand my overall learning pace and improve my study habits	Simple	Analytical
US10 NEW	Join virtual lessons	Language Learner	To be able to join and attend real-time virtual lessons with teachers and other users	I can practice conversation skills in real time and also receive feedback	Complex	Operational

2. Updated Conceptual Model

Based on feedback from TA on our Phase 1 Deliverable, we eliminated some entities and have simplified our conceptual model.



Assumptions:

1. Each vocabulary is only properly introduced with definition & example once. But it may be used in multiple places
2. A user can only be a learner or instructor not both
3. Hitting the daily goal is determined by XP earned that day (not minutes).

3. Relational Model

- **Primary keys** are in bold
- Foreign keys are underlined

Leaderboard(**lbid**, start_date, end_date, type)

Leaderboard_Spot(**lbid**, points, last_updated_date, last_updated_time, rank, lbid, lpid)

Lesson_Progress(**lpid**, streak_count, daily_xp, uid, cid)

Users(**uid**, email, valid)

Learner(**uid**, join_date, daily_goal, current_streak, highest_streak, last_goal_met_date)

Courses_taken(**uid,cid**)

Lesson(**lid**, name, difficulty_level, description, skill, cid)

Exercise(**eid**, name, type, accuracy, target_accuracy, num_attempts, score, lid, uid)

Virtual_Lesson(**vlid**, topic, language, start_time, end_time)

Attendance(**uid, vlid**, join_time, leave_time)

Prop(**pid**, name, type, uid)

Course(**cid**, language, is_active, successor_cid)

Sus_account (**flag_id**, reason, is_resolved, uid)

Lesson(**lid**, name, difficulty_level, description, skill)

Vocabulary(**vocab_id**, vocab, translation, example_sentence, difficulty_level)

Vocab_used_in_lesson(**lid, vocab_id**)

Game_Feature (**feature_id**, ffname, type, description, is_active)

Feature_used_in_lesson (**lid, feature_id**)

Design_metric (**metric_id**, completion_rate, avg_score, engagement_level, sample_size)

4. Functional Dependencies and Normalization

All functional dependencies and its relations are in BCNF. This means that each of the functional dependencies are good in regard to its relation.

A “good” functional dependency is when the key determines all other attributes in the relation.

Leaderboard(**lbid**, start_date, end_date, type)

- Lbid → start_date, end_date, type
- Lbid + → {lbid, start_date, end_date, type}

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Leaderboard_Spot(**lbsid**, points, last_updated_date, last_updated_time, rank, lbid, lpid)

- Lbsid → points, last_updated, rank, lbid, lpid
- Lbsid+ → {lbsid, points, last_updated, rank, lbid, lpid, start_date, end_date, type, streak_count, daily_xp, uid, cid, email, valid, language, is_active, successor_cid}

Lesson_Progress(**lpid**, streak_count, daily_xp, uid, cid)

- Lpid → streak_count, daily_xp, uid, cid, email, valid
- Lpid+ → {lpid, streak_count, daily_xp, uid, cid, language, is_active, successor_cid}

Users(**uid**, email, valid)

- uid → email, valid
- uid+ → {uid, email, valid}

Learner(**uid**, join_date, daily_goal, current_streak, highest_streak, last_goal_met_date)

- Uid → join_date, daily_goal, current_streak, highest_streak, last_goal_met_date
- Uid+ → {uid, email, valid, join_date, daily_goal, current_streak, highest_streak, last_goal_met_date}

Lesson(**lid**, name, difficulty_level, description, skill, cid)

- lid → name, difficulty_level, description, skill, cid
- lid+ → {lid, name, difficulty_level, description, skill, cid, language, is_active, successor_cid}

Exercise(**eid**, name, type, accuracy, target_accuracy, num_attempts, score, lid, uid)

- Eid → name, type, accuracy, target_accuracy, num_attempts, score, lid, uid

- Eid+ → {eid, name, type, accuracy, target_accuracy, num_attempts, score, lid, uid, lid, name, difficulty_level, description, skill, cid, language, is_active, successor_cid, email, valid}

Virtual_Lesson(**v**lid, topic, language, start_time, end_time)

- Vlid → topic, language, start_time, end_time
- Vlid+ → {topic, language, start_time, end_time}

Attendance(**u**id, **v**lid, join_time, leave_time)

- Uid, vlid → join_time, leave_time
- Uid, vlid+ → {uid, vlid, email, valid, topic, language, start_time, end_time, join_time, leave_time}

Prop(**p**id, name, type, uid)

- Pid → name, type, uid
- Pid+ → {pid, name, type, uid, email, valid}

Course(**c**id, language, is_active, successor_cid)

- cid → language, is_active, successor_cid
- cid+ → {cid, language, is_active, successor_cid}

Sus_account (**f**lag_id, reason, is_resolved, uid)

- Flag_id → uid, reason, is_resolved
- Flag_id+ → {flag_id, uid, reason, is_resolved, email, valid}

Vocabulary(**vocab_id**, vocab, translation, example_sentence, difficulty_level)

- Vocab_id → vocab, translation, example_sentence, difficulty_level
- Vocab_id+ → { vocab_id, vocab, translation, example_sentence, difficulty_level }

Game_Feature (**feature_id**, fname, type, description, is_active)

- feature_id → fname, type, description, is_active
- feature_id+ → {feature_id, fname, type, description, is_active}

Design_metric (**metric_id**, completion_rate, avg_score, engagement_level, sample_size)

- metric_id → completion_rate, avg_score, engagement_level, sample_size, feature_id
- metric_id+ → {metric_id, completion_rate, avg_score, engagement_level, sample_size, feature_id, fname, type, description, is_active}

5. Physical Model

